

$$\begin{aligned}
& y_u^{2112} \rightarrow \frac{1}{16\pi^2} \left(\tilde{\mu} y_e^{\text{pr}} y_u^{1112} \overline{a_e^{\text{pr}}} \text{LF}_{2,1,0} [m_l^{\text{P}}, m_e^{\text{r}}] - \tilde{\mu} y_e^{\text{pr}} y_u^{1112} \overline{a_e^{\text{pr}}} \text{LF}_{3,1,-1} [m_l^{\text{P}}, m_e^{\text{r}}] + \right. \\
& \frac{1}{2} y_u^{1112} (2 C_{\phi 2\phi 1} \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \overline{a_e^{\text{pr}}} (C_{\phi 2\phi 1} a_e^{\text{pr}} + 2 \tilde{\mu} y_e^{\text{pr}} (C_{\phi 1^2} + C_{\phi 2^2}))) \text{LF}_{3,1,0} [m_l^{\text{P}}, m_e^{\text{r}}] - \\
& \frac{3}{2} y_u^{1112} (2 C_{\phi 2\phi 1} \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \overline{a_e^{\text{pr}}} (C_{\phi 2\phi 1} a_e^{\text{pr}} + 2 \tilde{\mu} y_e^{\text{pr}} (C_{\phi 1^2} + C_{\phi 2^2}))) \text{LF}_{4,1,-1} [m_l^{\text{P}}, m_e^{\text{r}}] + \\
& y_u^{1112} (2 C_{\phi 2\phi 1} \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \overline{a_e^{\text{pr}}} (C_{\phi 2\phi 1} a_e^{\text{pr}} + 2 \tilde{\mu} y_e^{\text{pr}} (C_{\phi 1^2} + C_{\phi 2^2}))) \text{LF}_{5,1,-2} [m_l^{\text{P}}, m_e^{\text{r}}] + \\
& 3 \tilde{\mu} y_d^{\text{pr}} y_u^{1112} \overline{a_d^{\text{pr}}} \text{LF}_{2,1,0} [m_q^{\text{P}}, m_d^{\text{r}}] - 3 \tilde{\mu} y_d^{\text{pr}} y_u^{1112} \overline{a_d^{\text{pr}}} \text{LF}_{3,1,-1} [m_q^{\text{P}}, m_d^{\text{r}}] + \\
& \frac{3}{2} y_u^{1112} (2 C_{\phi 2\phi 1} \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} + \overline{a_d^{\text{pr}}} (C_{\phi 2\phi 1} a_d^{\text{pr}} + 2 \tilde{\mu} y_d^{\text{pr}} (C_{\phi 1^2} + C_{\phi 2^2}))) \text{LF}_{3,1,0} [m_q^{\text{P}}, m_d^{\text{r}}] - \\
& \frac{9}{2} y_u^{1112} (2 C_{\phi 2\phi 1} \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} + \overline{a_d^{\text{pr}}} (C_{\phi 2\phi 1} a_d^{\text{pr}} + 2 \tilde{\mu} y_d^{\text{pr}} (C_{\phi 1^2} + C_{\phi 2^2}))) \text{LF}_{4,1,-1} [m_q^{\text{P}}, m_d^{\text{r}}] + \\
& 3 y_u^{1112} (2 C_{\phi 2\phi 1} \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} + \overline{a_d^{\text{pr}}} (C_{\phi 2\phi 1} a_d^{\text{pr}} + 2 \tilde{\mu} y_d^{\text{pr}} (C_{\phi 1^2} + C_{\phi 2^2}))) \text{LF}_{5,1,-2} [m_q^{\text{P}}, m_d^{\text{r}}] + \\
& 3 \tilde{\mu} y_u^{\text{pr}} y_u^{1112} \overline{a_u^{\text{pr}}} \text{LF}_{2,1,0} [m_u^{\text{r}}, m_q^{\text{P}}] - 3 \tilde{\mu} y_u^{\text{pr}} y_u^{1112} \overline{a_u^{\text{pr}}} \text{LF}_{3,1,-1} [m_u^{\text{r}}, m_q^{\text{P}}] + \\
& \frac{3}{2} y_u^{1112} (C_{\phi 2\phi 1} \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + 2 \overline{a_u^{\text{pr}}} (C_{\phi 2\phi 1} a_u^{\text{pr}} + \tilde{\mu} y_u^{\text{pr}} (C_{\phi 1^2} + C_{\phi 2^2}))) \text{LF}_{3,1,0} [m_u^{\text{r}}, m_q^{\text{P}}] - \\
& \frac{9}{2} y_u^{1112} (C_{\phi 2\phi 1} \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + 2 \overline{a_u^{\text{pr}}} (C_{\phi 2\phi 1} a_u^{\text{pr}} + \tilde{\mu} y_u^{\text{pr}} (C_{\phi 1^2} + C_{\phi 2^2}))) \text{LF}_{4,1,-1} [m_u^{\text{r}}, m_q^{\text{P}}] + \\
& 3 y_u^{1112} (C_{\phi 2\phi 1} \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + 2 \overline{a_u^{\text{pr}}} (C_{\phi 2\phi 1} a_u^{\text{pr}} + \tilde{\mu} y_u^{\text{pr}} (C_{\phi 1^2} + C_{\phi 2^2}))) \text{LF}_{5,1,-2} [m_u^{\text{r}}, m_q^{\text{P}}] + \\
& m_1 \tilde{\mu} g_1^2 y_u^{1112} \text{LF}_{2,1,0} [\tilde{\mu}, m_1] - g_1^2 y_u^{1112} (3 C_{\phi 2\phi 1} + m_1 \tilde{\mu}) \text{LF}_{3,1,-1} [\tilde{\mu}, m_1] + \\
& m_1 \tilde{\mu} g_1^2 y_u^{1112} (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{3,1,0} [\tilde{\mu}, m_1] + 6 C_{\phi 2\phi 1} g_1^2 y_u^{1112} \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] - \\
& 3 m_1 \tilde{\mu} g_1^2 y_u^{1112} (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] - \\
& 3 C_{\phi 2\phi 1} g_1^2 y_u^{1112} \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] + 2 m_1 \tilde{\mu} g_1^2 y_u^{1112} (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] + \\
& 3 m_2 \tilde{\mu} g_2^2 y_u^{1112} \text{LF}_{2,1,0} [\tilde{\mu}, m_2] - 3 g_2^2 y_u^{1112} (3 C_{\phi 2\phi 1} + m_2 \tilde{\mu}) \text{LF}_{3,1,-1} [\tilde{\mu}, m_2] + \\
& 3 m_2 \tilde{\mu} g_2^2 y_u^{1112} (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{3,1,0} [\tilde{\mu}, m_2] + 18 C_{\phi 2\phi 1} g_2^2 y_u^{1112} \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] - \\
& 9 m_2 \tilde{\mu} g_2^2 y_u^{1112} (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] - 9 C_{\phi 2\phi 1} g_2^2 y_u^{1112} \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] + \\
& 6 m_2 \tilde{\mu} g_2^2 y_u^{1112} (C_{\phi 1^2} + C_{\phi 2^2}) \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] + \frac{2}{9} m_1 \tilde{\mu} g_1^2 y_u^{1112} \text{LF}_{1,1,1,0} [m_1, m_q^{i1}, m_u^{i2}] - \\
& \frac{1}{18} m_1 g_1^2 (C_{\phi 2\phi 1} a_u^{1112} + C_{\phi 2^2} \tilde{\mu} y_u^{1112}) \text{LF}_{2,2,1,-1} [m_1, m_q^{i1}, m_u^{i2}] + \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_u^{1112} \text{LF}_{1,1,1,0} [m_1, m_q^{i1}, \tilde{\mu}] - \frac{1}{18} m_1 g_1^2 (C_{\phi 2\phi 1} a_u^{1112} + C_{\phi 2^2} \tilde{\mu} y_u^{1112}) \\
& \text{LF}_{2,2,1,-1} [m_1, m_u^{i2}, m_q^{i1}] - \frac{2}{3} m_1 \tilde{\mu} g_1^2 y_u^{1112} \text{LF}_{1,1,1,0} [m_1, m_u^{i2}, \tilde{\mu}] - \\
& \frac{1}{12} C_{\phi 2\phi 1} g_1^2 y_u^{1112} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_q^{i1}] + \frac{1}{12} m_1 C_{\phi 2^2} \tilde{\mu} g_1^2 y_u^{1112} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{i1}] + \\
& \frac{1}{3} C_{\phi 2\phi 1} g_1^2 y_u^{1112} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_u^{i2}] - \frac{1}{3} m_1 C_{\phi 2^2} \tilde{\mu} g_1^2 y_u^{1112} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^{i2}] - \\
& \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_u^{1112} \text{LF}_{1,1,1,0} [m_2, m_q^{i1}, \tilde{\mu}] + \frac{3}{4} C_{\phi 2\phi 1} g_2^2 y_u^{1112} \text{LF}_{2,2,1,-2} [m_2, \tilde{\mu}, m_q^{i1}] - \\
& \frac{3}{4} m_2 C_{\phi 2^2} \tilde{\mu} g_2^2 y_u^{1112} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{i1}] + \frac{8}{3} m_3 \tilde{\mu} g_3^2 y_u^{1112} \text{LF}_{1,1,1,0} [m_3, m_q^{i1}, m_u^{i2}] - \\
& \frac{2}{3} m_3 g_3^2 (C_{\phi 2\phi 1} a_u^{1112} + C_{\phi 2^2} \tilde{\mu} y_u^{1112}) \text{LF}_{2,2,1,-1} [m_3, m_q^{i1}, m_u^{i2}] - \\
& \frac{2}{3} m_3 g_3^2 (C_{\phi 2\phi 1} a_u^{1112} + C_{\phi 2^2} \tilde{\mu} y_u^{1112}) \text{LF}_{2,2,1,-1} [m_3, m_u^{i2}, m_q^{i1}] + \\
& \tilde{\mu} y_d^{11r} y_u^{\text{pi}2} \overline{a_d^{\text{pr}}} \text{LF}_{1,1,1,0} [m_d^{\text{r}}, m_q^{\text{P}}, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} y_d^{11r} y_u^{\text{pi}2} (C_{\phi 2^2} \overline{a_d^{\text{pr}}} + C_{\phi 2\phi 1} \tilde{\mu} \overline{y_d^{\text{pr}}}) \\
& \text{LF}_{2,1,1,0} [m_d^{\text{r}}, m_q^{\text{P}}, \tilde{\mu}] - \frac{1}{2} \tilde{\mu} y_d^{11r} y_u^{\text{pi}2} (C_{\phi 2^2} \overline{a_d^{\text{pr}}} + C_{\phi 2\phi 1} \tilde{\mu} \overline{y_d^{\text{pr}}}) \text{LF}_{3,1,1,-1} [m_d^{\text{r$$